

Facial Expression Recognition Using Edge AI

(7 Universal Emotions Based System)

Problem Statement

Understanding human emotions through facial expressions is a critical component of intelligent human-computer interaction. Most existing systems rely on cloud-based deep learning models, which introduce latency, require high computational resources, and raise privacy concerns.

This project aims to design a **Facial Expression Recognition system using Edge AI principles**, capable of identifying the **7 universal human emotions** in real-time while minimizing dependency on cloud processing and preserving user privacy.

Key Objectives Include

- To detect and classify **7 universal facial expressions**:
 - Happiness
 - Sadness
 - Fear
 - Disgust
 - Anger
 - Contempt
 - Surprise
 - To simulate an **Edge AI pipeline for emotion recognition**
 - To extract **facial features** such as:
 - Eyebrow position
 - Mouth shape
 - Eye openness
 - To implement a **lightweight classification system** (rule-based / hybrid AI)
 - To provide **real-time predictions with confidence scores**
 - To enable **interactive system responses based on emotions**
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Features

Face & Feature Detection (Hybrid System)

- Camera input processed via external system (e.g., Python + OpenCV)
 - Detects:
 - Face landmarks
 - Eye openness
 - Mouth curvature
 - Eyebrow positioning
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Feature Extraction Layer

Converts raw facial data into usable signals:

- Smile intensity → Happiness
- Eyebrow angle → Anger / Fear

- Mouth openness → Surprise
- Nose wrinkle → Disgust
- Asymmetry → Contempt

Emotion Classification Engine

Maps features → emotion labels:

Emotion	Key Indicators
Happiness	Raised cheeks, smiling lips
Sadness	Drooping mouth, inner eyebrow lift
Fear	Wide eyes, stretched mouth
Disgust	Wrinkled nose
Anger	Lowered brows, tight lips
Contempt	One-sided lip raise
Surprise	Raised brows, open mouth

AI Simulation Layer

- Confidence scoring (simulated probabilities)
- Noise handling via smoothing
- Priority-based classification (avoid conflicts)

Real-Time Output

System reacts via:

- LEDs (emotion indication)
- Servo motors (expression-based movement)
- Serial monitor dashboard

Performance Tracking

- Tracks prediction accuracy
- Compares expected vs predicted emotion
- Displays live system metrics

Goals

- To build a **real-time emotion-aware system**
- To simulate **Edge AI-based facial recognition pipelines**
- To explore **human emotion modeling in machines**
- To create a **touchless, emotion-driven interface**
- To prepare for **TinyML + embedded vision systems**

Benefits

- **Emotion-Aware Interaction** – Systems respond to user feelings
- **Low Latency** – Faster than cloud-based solutions
- **Privacy-Friendly** – Facial data stays local
- **High Research Value** – Aligns with AI, HCI, and robotics
- **Scalable** – Can upgrade to real ML models (CNNs, TensorFlow Lite)

- **Real-World Applications:**
 - Smart assistants
 - Healthcare monitoring
 - Driver alert systems
 - Adaptive gaming
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Conclusion

This project demonstrates a **Facial Expression Recognition system based on the 7 universal human emotions**, implemented using Edge AI principles. By combining facial feature extraction, classification logic, and real-time system response, it simulates how intelligent systems can interpret and react to human emotions.

Although full facial recognition requires higher computational power than microcontrollers like Arduino, the hybrid approach (external vision processing + embedded response system) successfully models a **practical Edge AI architecture**.

This makes the project a strong foundation for future work in:

- **Affective Computing**
- **TinyML-based vision systems**
- **Human-centered AI interfaces**